Position, distance, displacement

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish a location, a total path length, and a change in position.

SUCCESS CRITERIA - I can...

1. Label the origin, start, turn and finish.
2. Calculate path distance and signed displacement separately.
3. Explain why position and displacement can differ.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Positions are 15, 10, 5 m at 0, 2, 4 s. Predict position at 7 s. Give the rate, an assumption and a future test.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta x = x_f - x_i$; distance = sum of path lengths. Choose an origin and positive direction. i = initial; f = final.

Worked reasoning: A walker starts at 2 m, reaches 6 m, then ends at 4 m. Distance=4+2=6 m. Displacement=4-2=2 m right.

Your turn: The walker goes from the 2 m floor mark to the 6 m mark, then back to 4 m. On the number line, draw the two legs above the axis and a separate start-to-finish arrow below it. Label each length.

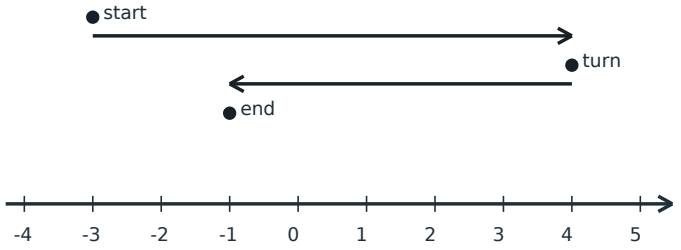


Position (m); right is positive

Marzano check - circle: ① ② ③

03 Track a signed route

A robot moves along an east-west test track. East is positive. It starts at -3 m, reaches +4 m, then stops at -1 m. Use the route to find total distance and displacement.



Position (m); right is positive

Marzano check - circle: ① ② ③

04 Change the coordinate convention

For the route in 3, reverse the positive direction while keeping the origin fixed. Give the new endpoint coordinates and displacement. What stays unchanged?

Marzano check - circle: ① ② ③

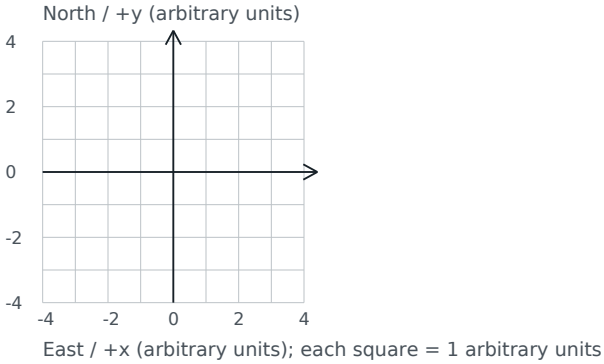
Position, distance, displacement

antocciphysics

Continue down each column.

05 A closed path

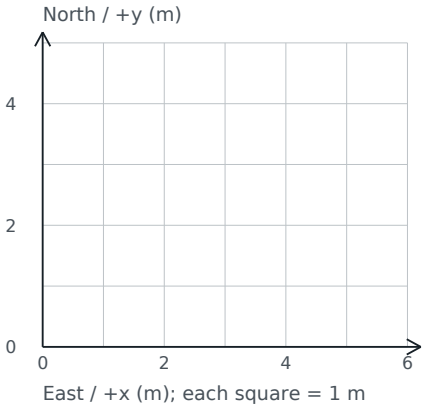
A runner completes one 400 m lap of a circular track, then runs another 100 m. Sketch the full trip and mark the final point. Is the displacement magnitude 500 m? Explain how the track geometry determines the start-to-finish distance.



Marzano check - circle: ① ② ③

06 Different paths, same endpoints

A robot must travel from (0, 0) m to (4, 3) m. Draw two routes on the grid: one using only east/north legs and one direct route. Find each path length and the displacement magnitude both routes share.



Marzano check - circle: ① ② ③

07 Zero and nonzero

Can displacement be zero while distance is nonzero? Can distance be zero while displacement is nonzero? Justify both answers.

Marzano check - circle: ① ② ③

08 Review: connect an earlier idea

A fictional observing report records 3 images of a moving object but gives no distance or size calibration. State a supported observation and explain why a precise impact prediction is unsupported.
Separate what was observed from what can be inferred.

Observation	Supported claim	Missing information
___	___	___
___	___	___

Marzano check - circle: ① ② ③

09 Exit check

Start at +5 m, walk to -2 m, and finish at +1 m. Calculate distance and displacement. Explain why neither equals final position.

Marzano check - circle: ① ② ③

10 STRETCH: test the model

In one dimension, start at -2 m and end at +3 m with total distance 11 m. Find two possible single-turn positions and draw both routes.



Position (m); right is positive

Marzano check - circle: ① ② ③